Homework 1

by Filip Rehburg

This homework is worth 100 points in total. This homework is **individual** (so, **no collaboration**!). Please **type in capital letters your full name**. Your answers should be **typed** as PDF, except for the drawings, which can scanned hand-drawings integrated with your pdf.

1. *(35 points)* A non-standard version of the so-called *Singapore Problem* goes as follows:

   Albert and Bernard just met Cheryl. "When is your birthday?" Albert asks Cheryl. Cheryl thinks a second and says, "Boys, I'm not going to tell you this, but I'll give you some clues, see how smart you are. Then I'll go out on a date with whoever of you finds the answer first". Then she writes down a list of 10 dates:

   May 15, May 16, May 19, June 14, July 17, July 18, August 14, August 15, August 17, September 14, September 16.

   "My birthday is one of these," she tells them. "Now I am going to tell to one of you the month (and only the month) of my birthday, and tell to the other the day (and only the day) of my birthday." Then Cheryl whispers in Albert's ear the month (and only the month) of her birthday. To Bernard, she whispers the day (and only the day).

   "Do you think that Albert can figure our my exact birthday now?" she asks Bernard.

   Bernard: "Huh! I know for sure that Albert doesn't know when it is!"

   Albert (after hearing Bernard's statement): "Yes, but I also I know that right now poor Bernard doesn't know it either!"

   Bernard (after listening to Albert(: "Well, well, guess what: now I know your birthday, but… I also know that Albert knows it too! So hmm, you won't go out with either of us then?! Do you have another test for us? Or shall we throw a fair coin??""

   The problem is: *When is Cheryl's birthday?*

   (a) *(7 points)* First, start by representing (drawing) the epistemic situation immediately after Cheryl gives the boys their pieces of information (but before she starts questioning them). Represent all the facts, as well as each agent's knowledge in this situation, using a Kripke model M with three agents $\mathcal{A} = \{\text{Albert}, \text{Bernard}, \text{Cheryl}\}$ and ten atomic propositions

   $$\text{Prop} = \{\text{May}, \text{June}, \text{July}, \text{August}, \text{September}, 14, 15, 16, 17, 18, 19\}, \tag{1}$$

   where the first four indicate "Cheryl's birthday is in month x''" and the others indicate "Cheryl's birthday is in day y". (You may disregard all the information about going out on a date, that is just for fun.) **Draw the model (as a graph)**, don't just describe it in words or formulas!

Answer to Exercise 1. (a)

Let

$\mathcal{A} = \{a, b, c\}$

$\text{Prop}_M = \{M, \text{Jn}, \text{Jl}, A, S\}$

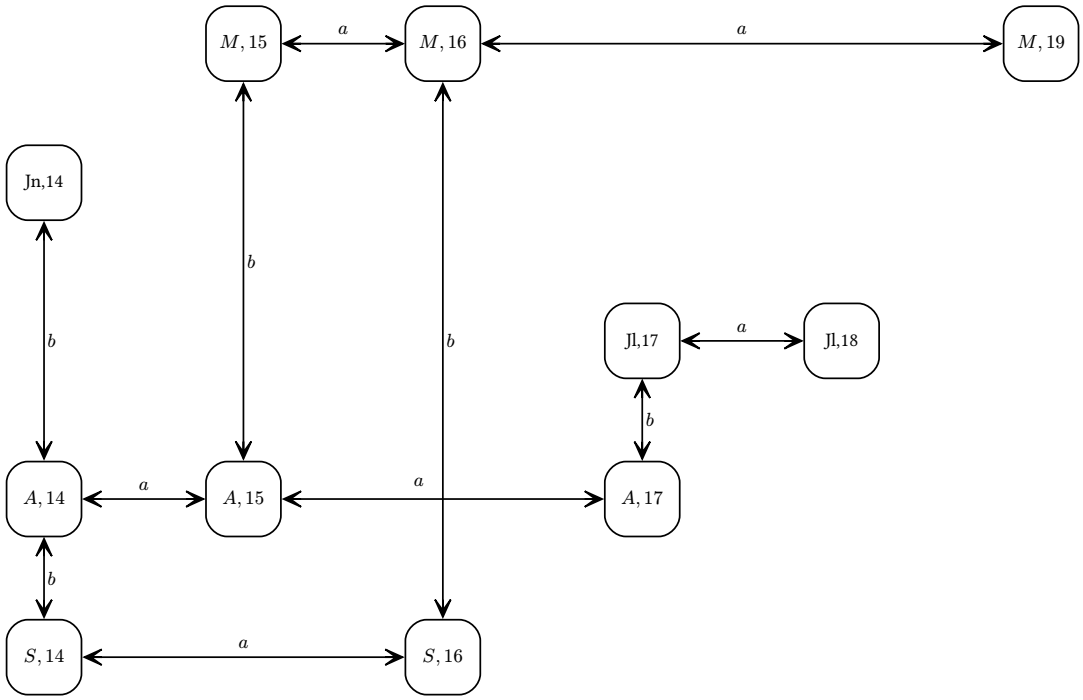
$\text{Prop}_D = \{14, 15, 16, 17, 18, 19\}$

$\text{Prop} = \text{Prop}_M \cup \text{Prop}_D$

$S = \{(M, 15), (M, 16), (M, 19), (\text{Jn}, 14), (\text{Jl}, 17), (\text{Jl}, 18), (A, 14), (A, 15), (A, 17), (S, 14), (S$

It is common knowledge that a knows the birthday month and b knows the date. $Ck\left(\bigvee_{m \in \text{Prop}_M} \text{K}_a m\right) \wedge Ck_{a,b,c,}\left(\bigvee_{d \in \text{Prop}_D} \text{K}_b d\right)$.

The following is a graph describing epistemic model $\mathbf{S} = \left(S, \left\{\underset{\alpha}{\sim}\right\}_{\alpha \in \mathcal{A}}, \|\cdot\|\right)$. Since this is an epistemic **S5** model, all relations are reflexive, transitive, and thus symmetric. Reflexive and transitive arrows are omitted for visual clarity. Since Cheryl c knows her own birthday, all worlds form their own equivalence class under $\underset{c}{\sim}$ $(\forall s = (m, d), t = (m', d') \in S : s \underset{c}{\sim} t \Leftrightarrow (m = m' \wedge d = d'))$.



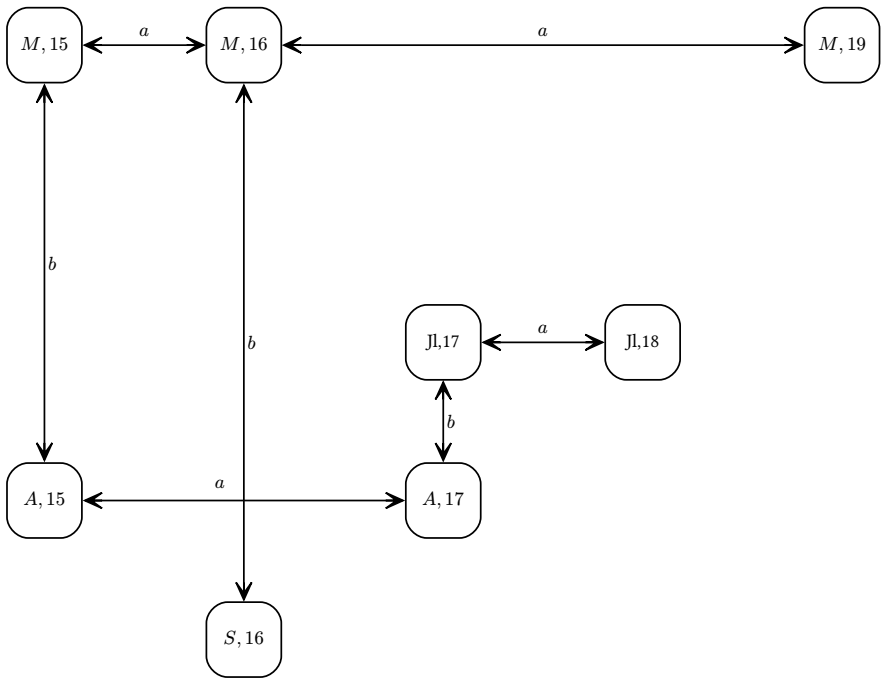
(b) *(7 points)* As for the Muddy Children, write down an epistemic sentence encoding Bernard's first announcement (that he knows Albert doesn't know when her birthday is). As for the Muddy Children, interpret Bernard's first announcement as a truthful public announcement, and represent (draw) the updated model M' after this update.

Answer to Exercise 1. (b)

Bernards first announcement encoded as an epistemic sentence is:

$$\varphi_1 = \mathrm{K}_b \left(\bigwedge_{(m,d)\in S} (\neg \mathrm{K}_a (m \wedge d)) \right) \tag{2}$$

The updated model $\mathbf{S}'$:



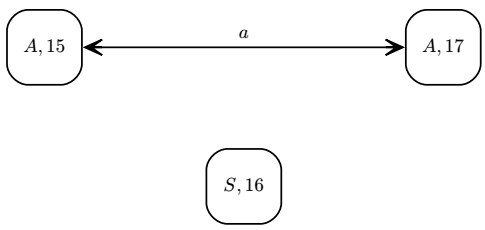
(c) *(7 points)* Do the same as in previous part for Albert's first announcement (that now he also knows that Bernard doesn't know Cheryl's birthday): encode his statement as an epistemic sentence, and represent (draw) the updated model M'' after this new update.

Answer to Exercise 1. (c)

Albert's first announcement encoded as an epistemic sentence is:

$$\psi_1 = \left(\mathrm{K}_a \bigwedge_{(m,d)\in S} (\neg (m \wedge d)) \right) \wedge \left(\mathrm{K}_a \bigwedge_{(m,d)\in S} (\neg \mathrm{K}_b (m \wedge d)) \right) \tag{3}$$

The updated model $\mathbf{S}''$:



(d) *(7 points)* Do the same for Bernard's second announcement (that now he knows the birthday, but that he also knows that Albert knows the birthday as well), computing the updated model M'' Use this to solve the puzzle: when is Cheryl's birthday?

Answer to Exercise 1. (d)

Bernard's second announcement encoded as an epistemic sentence is:

$$\varphi_2 = \left(\bigvee_{(m,d)\in S} \mathrm{K}_b(m \wedge d) \wedge \mathrm{K}_b \left(\bigvee_{(m,d)\in S} \mathrm{K}_a(m \wedge d) \right) \right) \tag{4}$$

From $\mathbf{S}''$ we can deduce that c's birthday is September the 16th ($s_\star = (S, 16)$).

(e) *(7 points)* Write a sentence in the language of Public Announcement Logic (PAL) saying that: "after Bernard's first announcement, followed by Albert's first announcement, Bernard knows Cheryl's birthday and he also knows that Albert knows it".

Answer to Exercise 1. (e)

$![\varphi_1]![\psi_1]\varphi_2$

2. *(35 points)* Alice and Bob have each some positive natural number $n_a, n_b \in \{1, 2, ..., 12\}$ written on their forehead. It is common knowledge that
   1. each of them can see the other's number (Alice can see n_b and Bob can see n_a), but neither of them can see his/her number;
   2. the two children are perfect logicians.
   3. both numbers are no larger than 12 (i.e. $1 \leq n_a, n_b \leq 12$);
   4. the two numbers are related by the function $g : \mathbb{N} \to \mathbb{N}$ given by: $g(n) = 1$, if $n = 2^k$ for some $k > 0$; $g(n) = n + 2$, if n is odd; $g(n) = n - 2$ if n is even but not a power of 2.

$$g(n) = \begin{cases} 1 \text{ if } n = 2^k \text{ for some } k > 0 \\ n + 2 \text{ if n is odd} \\ n - 2 \text{ if } n \text{ is even, but not a power of two} \end{cases} \tag{5}$$

   4. So it is common knowledge that either $n_a = g(n_b)$ or else $n_b = g(n_a)$.

The Father asks them, repeatedly: "Do you know your own number?". The two are supposed to answer truthfully, publicly, simultaneously (without any other communication). They both answer "I don't know" to the first 3 questions, after which Alice answers "Yes, now I know my number" to the 4th question (while Bob still answers "I don't know").

(a) *(10 points)* What is Alice's number?

Answer to Exercise 2 (a)

a has the number 3 written on their forehead.

$$n_a = 3 \tag{6}$$

(b) *(5 points)* Will Bob ever know his number (without looking in the mirror or being told the number by Alice)? If so, when will he answer "I know my number"?

Answer to Exercise 2 (a)

b will never know his number (without looking in the mirror or being told the number by a.

(c) *(20 points)* Prove your conclusions in (a) and (b) semantically, by drawing the initial epistemic model, then applying repeated updates with the (semantic information conveyed by the) children's answers. Draw each intermediary model, and explain your conclusions.

Answer to Exercise 2 (c): Initial Situation

All values of $g(n)$ computed:

$n = 1, g(n) = 3 \quad n = 2, g(n) = 1$

$n = 3, g(n) = 5 \quad n = 4, g(n) = 1$

$n = 5, g(n) = 7 \quad n = 6, g(n) = 4$

$n = 7, g(n) = 9 \quad n = 8, g(n) = 1(7)$

$n = 9, g(n) = 11 \quad n = 10, g(n) = 8$

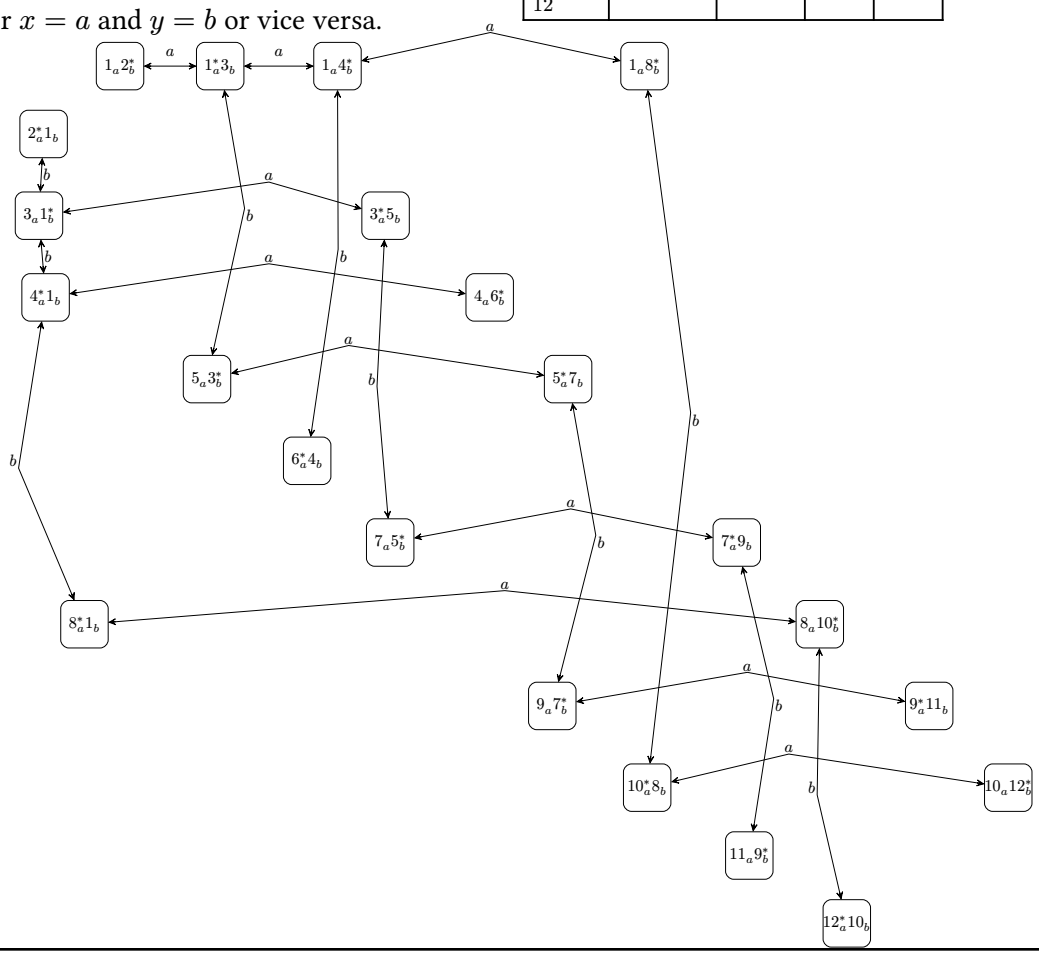
$n = 11, g(n) = 13 > 12 \lightning$

$n = 12, g(n) = 10$

Denote by $G_i(n_y)$ the set of possible values for n_x upon seeing n_y under g after announcement round i, where either $x = a$ and $y = b$ or vice versa.

	possible values for n_x			
x sees	$i = 0$	$i = 1$	$i = 2$	$i = 3$
$n_y = 1$	$\{2, 3, 4, 8\}$	$\{3, 4, 8\}$	$\{3, 8\}$	$\{3\}$
$n_y = 2$	$\{1\}$	$\emptyset$	$\emptyset$	$\emptyset$
$n_y = 3$	$\{1, 5\}$	$\{1, 5\}$	$\{1, 5\}$	$\{1, 5\}$
$n_y = 4$	$\{1, 6\}$	$\{1\}$	$\emptyset$	$\emptyset$
$n_y = 5$	$\{3, 7\}$	$\{3, 7\}$	$\{3\}$	$\{3\}$
$n_y = 6$	$\{4\}$	$\emptyset$	$\emptyset$	$\emptyset$
$n_y = 7$	$\{5, 9\}$	$\{5, 9\}$	$\{5\}$	$\emptyset$
$n_y = 8$	$\{1, 10\}$	$\{1, 10\}$	$\{1\}$	$\emptyset$
$n_y = 9$	$\{7, 11\}$	$\{7\}$	$\emptyset$	$\emptyset$
$n_y = 10$	$\{8, 12\}$	$\{8\}$	$\emptyset$	$\emptyset$
$n_y = 11$	$\{9\}$	$\emptyset$	$\emptyset$	$\emptyset$
$n_y = 12$	$\{10\}$	$\emptyset$	$\emptyset$	$\emptyset$

S:



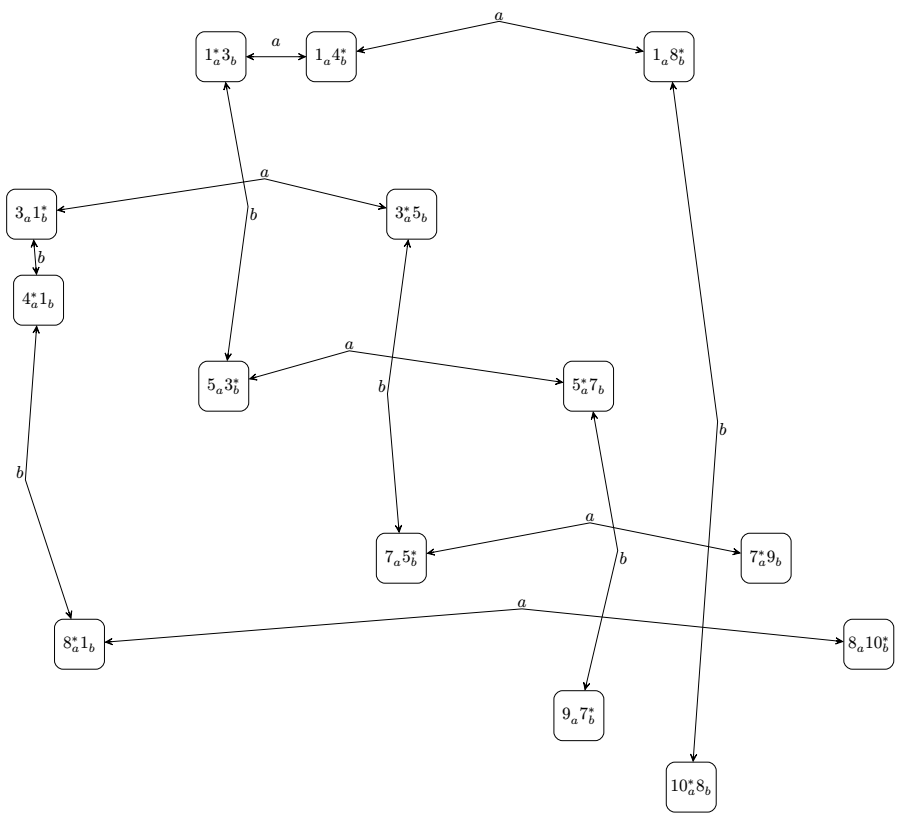
Answer to Exercise 2 (c) continued (i): After the first announcements of ignorance.

Eliminations:

(**ER1**) Elimination rule 1: Whenever x sees $n_y = u$ and $G_i(u)$ is a singleton $\{o\}$, then x knows their number is o. If they continue to pledge ignorance, o cannot be n_x. (Note the symmetry in the updates while both answer that they don't know).

1. $(1, 2)$ and $(2, 1)$: $G_0(2) = \{1\}$. Thus upon seeing $n_y = 2$ x knows $n_x = 1$. But the continued pledge of ignorance eliminates these worlds.
2. $(6, 4)$ and $(4, 6)$. following the same logic.
3. $(11, 9)$ and $(9, 11)$ following the same logic.
4. $(12, 10)$ and $(10, 12)$ following the same logic.

$\mathbf{S}'$:



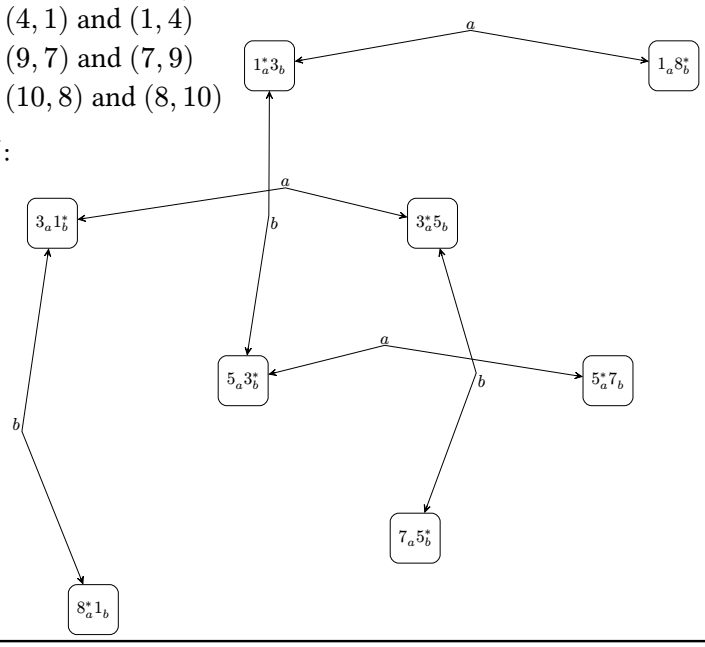
Answer to Exercise 2 (c) continued (ii): After the second announcements of ignorance.

Eliminations:

(**ER2**) Elimination rule 2: After eliminating possible worlds, the set of possible values $G_i(n_y)$ for n_x upon seeing n_y shrinks, allowing the iteration of (**ER1**).

5. $(4, 1)$ and $(1, 4)$
6. $(9, 7)$ and $(7, 9)$
7. $(10, 8)$ and $(8, 10)$

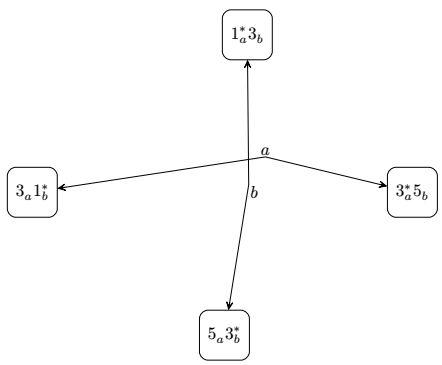
$\mathbf{S}''$:



Answer to Exercise 2 (c) continued (iii): After the third announcements of ignorance.

Eliminations:

8. $(8, 1)$ and $(1, 8)$
9. $(7, 5)$ and $(5, 7)$



Iteration 4: a knows, b does not know

How can a know their number? Scenarios:

1. a sees $n_b = 1$: Then a would know $n_a = 3$.
2. a sees $n_b = 3$: $G_3(3) = \{1, 5\}$, so a couldn't truthfully announce that they know n_a, since there would be 2 choices left.
3. a sees $n_b = 5$: Then a would know $n_a = 3$.

Thus, we rule out scenario 2. In either scenario 1 or 3, $n_a = 3$.

3. *(30 points)*

(a) *(15 points)* Is the following formula valid or not on epistemic models with two agents (a and b)?

$$\mathrm{K}_a\Big(Dk_{a,b}(p) \vee Dk_{a,b}(\neg p)\Big) \Rightarrow Dk_{a,b}(\mathrm{K}_a(p) \vee \mathrm{K}_a(\neg p)) \tag{8}$$

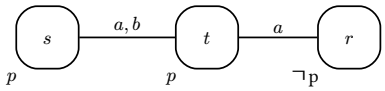
If your answer is yes, then give a semantic argument for this (by reasoning on possible worlds in any arbitrary epistemic model, using the semantic clauses for K, $Dk_{a,b}$, and propositional connectives). **If you are answer is no, then provide a counterexample** (by drawing an epistemic model, and presenting some world in that model, and showing that the premise is true and the conclusion false in that world).

NOTE: Recall that epistemic models are Kripke models in which all relations are equivalence relations, K_x represents knowledge by a specific agent (indicated by the subscript x), while $Dk_{a,b}$ represents distributed knowledge (in the group composed of agents a and b).

Answer to Exercise 3 (a)

Let $\mathcal{A} = \{a, b\}$.

Countermodel $\mathbf{S} = \Big(S, \big\{\underset{\alpha}{\sim}\big\}_{\alpha\in\mathcal{A}}, \|\cdot\|\Big)$: (Reflexive and transitive connections not drawn for simplicity)



Proof Premise True:

The equivalence classes for the agents are:
- $[s]_a = \{s, t, r\}$
- $[s]_b = \{s, t\}$ and $[r]_b = \{r\}$

The equivalence classes for distributed knowledge ($Dk_{a,b}$) are the intersections of the agents' classes:
- $[s]_{a,b} = \{s, t\}$
- $[r]_{a,b} = \{r\}$

1. Since $\forall x, y \in \{s, t\} : x \underset{\mathbf{S}}{\models} p$ and $[s]_{a,b} = \{s, t\}$, we deduce that $\forall x \in [s]_{a,b} : x \underset{\mathbf{S}}{\models} Dk_{a,b}(p)$.
2. From $r \underset{\mathbf{S}}{\models} \neg p$ and $[r]_{a,b} = \{r\}$, we get $r \underset{\mathbf{S}}{\models} Dk_{a,b}(\neg p)$.

By propositional weakening, we conclude $\forall x \in \{s, t, r\} : x \underset{\mathbf{S}}{\models} \Big(Dk_{a,b}(p) \vee Dk_{a,b}(p)\Big)$.

Note $[s]_a = \{r, s, t\} = S$. Since the disjunction holds in all worlds $\forall x \in S : Dk_{a,b}(p) \vee Dk_{a,b}(p)$, we conclude $\forall x \in S : x \underset{\mathbf{S}}{\models} \mathrm{K}_a\Big(Dk_{a,b}(p) \vee Dk_{a,b}(p)\Big)$.

Proof conclusion false:

In an S5 epistemic model, $x \models \mathrm{K}_a(\varphi)$ iff φ is true in all worlds in $[x]_a$.
- $s \not\models \mathrm{K}_a(p)$ because $r \in [s]_a$ and $r \models \neg p$.
- $s \not\models \mathrm{K}_a(p)$ because $s \in [s]_a$ and $s \models p$.

Since $\mathrm{K}_a(p) \wedge \neg\mathrm{K}_a(\neg p)$ holds for all nodes, $Dk_{a,b}(\mathrm{K}_a(p) \vee \mathrm{K}_a(\neg p))$ cannot hold anywhere.

□

(b) *(15 points)* Is the converse formula valid or not on epistemic models with two agents (a and b)?

$$Dk_{a,b}(\mathrm{K}_a(p) \vee \mathrm{K}_a(\neg p)) \Rightarrow \mathrm{K}_a\Big(Dk_{a,b}(p) \vee Dk_{a,b}(\neg p)\Big) \tag{9}$$

As before, **if your answer is yes then give a semantic argument for this, while you are answer is no then provide a counterexample.**

Answer to Exercise 3 (b)

Take an arbitrary epistemic model $\mathbf{S}$ with two agents, w.l.o.g.: $\mathcal{A} = \{a, b\}$. Assume $\underset{\mathbf{S}}{\models} Dk_{a,b}(\mathrm{K}_a(p) \vee \mathrm{K}_a(\neg p))$. Take an arbitrary $s \in \mathbf{S}$. Denote its equivalence class under distributed knowledge as $[s]_{a,b}$.

Note that at most one term in the premise disjunction can hold at a time, since $\mathrm{K}_a(p) \wedge \mathrm{K}_a(\neg p)$ is a contradiction. Thus, w.l.o.g., assume $s \underset{\mathbf{S}}{\models} Dk_{a,b}(\mathrm{K}_a(p))$.

Take an arbitrary t s.t. $s \underset{a}{\sim} t$. Since $s \underset{\mathbf{S}}{\models} Dk_{a,b}(\mathrm{K}_a(p))$, there must be a $w \in [s]_{a,b}$: $w \underset{\mathbf{S}}{\models} \mathrm{K}_a(p)$ (Note: $s \underset{a}{\sim} w$). Thus $\forall x \in [s]_{a,b} : x \underset{\mathbf{S}}{\models} p$, in particular s and w. From this we infer that $Dk_{a,b}(p)$ holds across $[s]_{a,b}$. By the Kripke semantics of distributed knowledge (intersection) $[s]_{a,b} \subseteq [s]_a$. Therefore, $w \underset{a}{\sim} t$ and from $w \underset{\mathbf{S}}{\models} \mathrm{K}_a(p)$, we get $t \underset{\mathbf{S}}{\models} p$ and this holds across $[s]_a$. $\forall x \in [s]_a : x \underset{\mathbf{S}}{\models} Dk_{a,b}(p)$. By propositional weakening $Dk_{a,b}(p) \vee Dk_{a,b}(\neg p)$ holds everywhere in $[s]_a$. By the semantics of knowledge, $\mathrm{K}_a\Big(Dk_{a,b}(p) \vee Dk_{a,b}(\neg p)\Big)$ holds across $[s]_a$, in particular at s, proving our goal.

$\square$